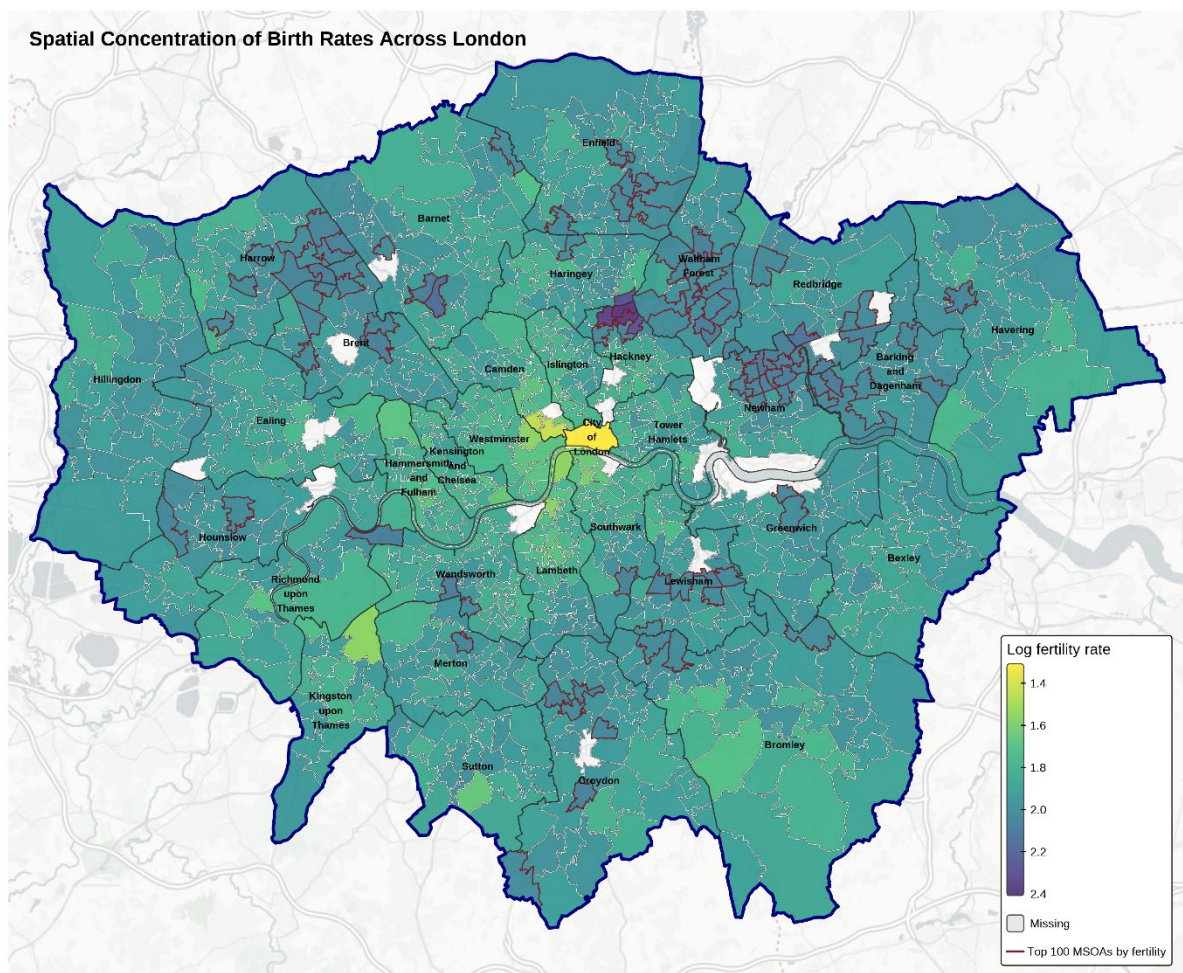


Where Families Still Form Under Housing Constraints in London

Name:

Family formation in high-density global cities is often portrayed as a struggle, mostly due to rising housing costs and the dominance of small dwellings. In London, these pressures are particularly High. Where fertility has fallen sharply to **1.35 children per woman** in 2024 [@ONS_Birth2024].

Rather than focusing on the decline in birth rates, we might ask: **where in London does early family formation remain viable?**



This report examines how housing structure and cost shape the geography of young families across London. Rather than if rising prices and shrinking dwellings inevitably push families outward towards the periphery, the analysis identifies where this pattern holds within London—and where it doesn't.

Across the city, neighborhoods with many small dwellings and high housing costs tend to record lower birth rates, according to the Assembly report. The report notes that births in 2023 were 20% lower than in 2012, driven by an "unfavorable environment" created by housing and childcare costs. [@ASSM_LDN]. Yet this is not universal. Some areas maintain high fertility despite intense housing pressure, suggesting that family formation persists even under conditions widely assumed to suppress it.

Measuring Housing Pressure

To understand these patterns, the analysis focuses on two structural features of the housing environment.

Housing form

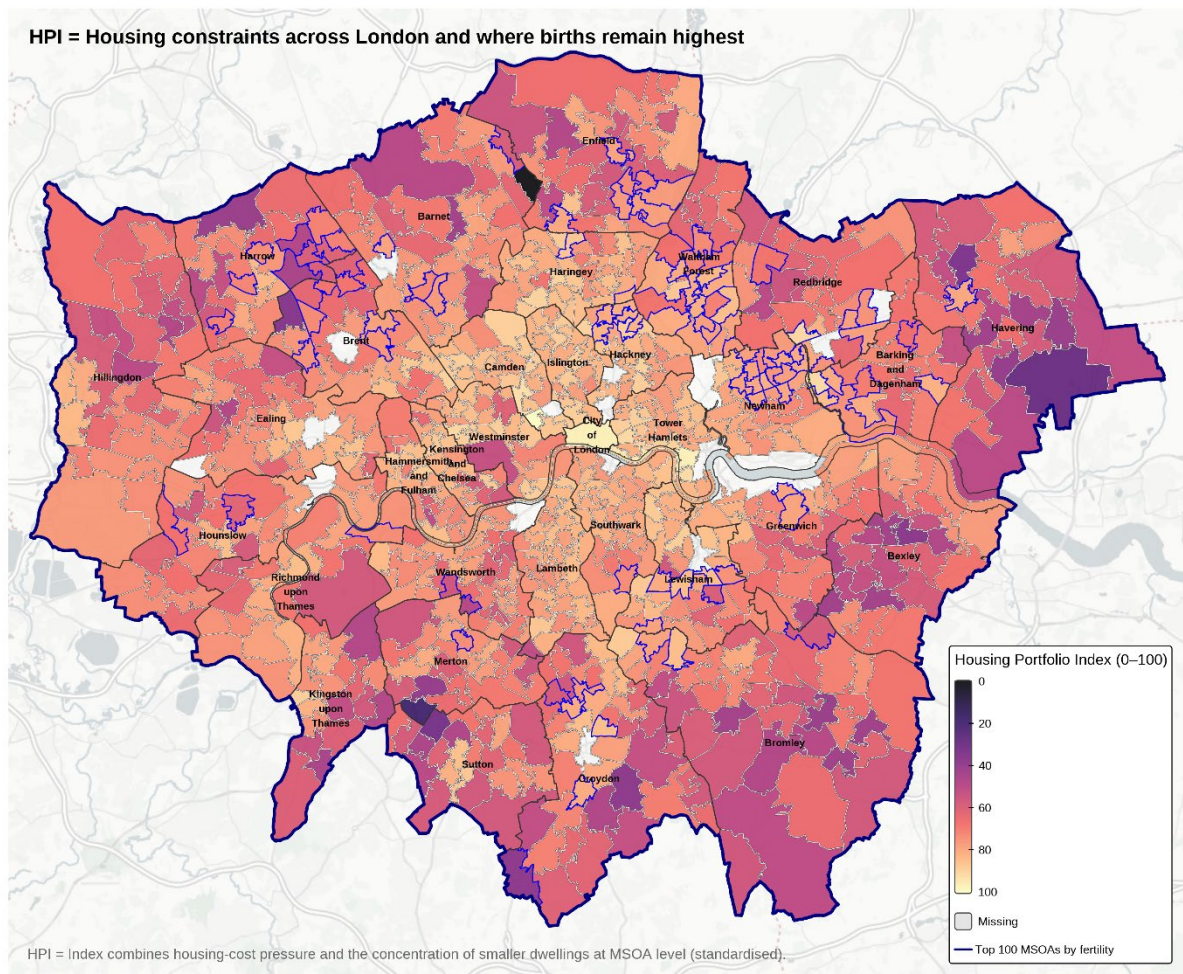
This captures the size and type of dwellings. One-bedroom homes, while not a perfect proxy, signal a housing stock poorly aligned with family needs. Their dominance indicates neighborhoods where space for family life is structurally limited and where they are more flexible. [@Data_Housing]

Housing cost pressure

Defined as the gap between disposable income before and after housing costs, this measure reflects the financial burden imposed by the local market. It captures affordability more effectively than income alone. Yet it's imperfect, as it aggregates at the neighborhood level, generalizes to a single household without accounting for household-size variation, and lacks information on gender and demographic factors. [@Data_H_Price]

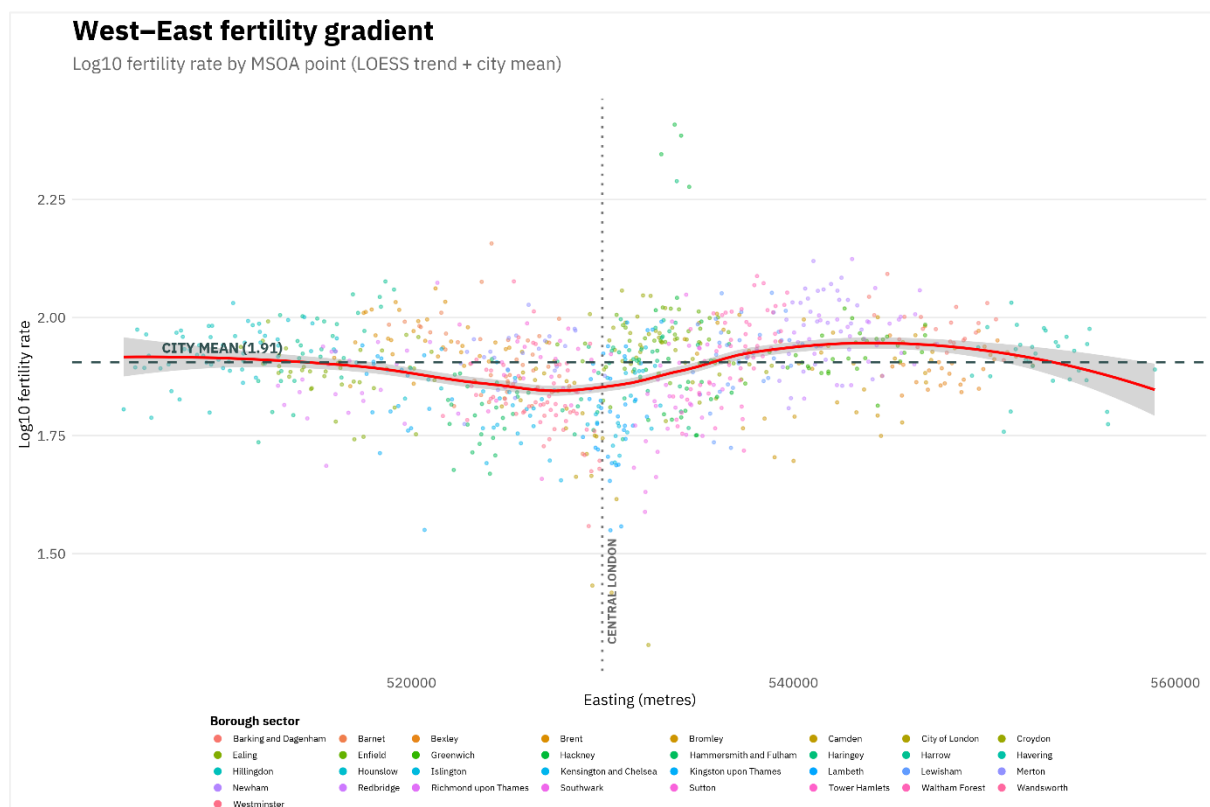
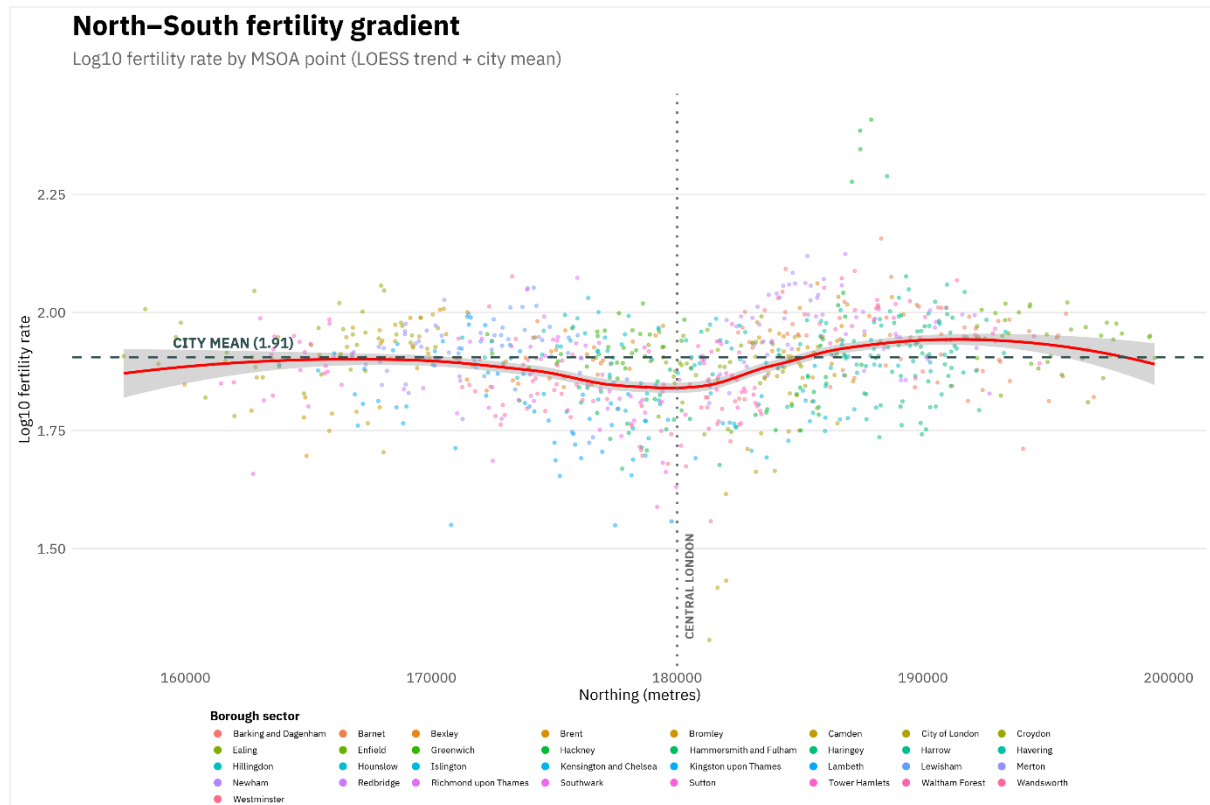
Combined, these measures form the **Housing Portfolio Index (HPI)**, where higher values indicate neighborhoods that are both expensive for the average household and

structurally constrained.



Fertility Across the City

Fertility rates [`@Birth_rate`], measured as births per 1,000 women aged 15–55 and averaged across 2019–2021, show a familiar pattern: lower rates in much of inner London and higher rates in outer boroughs, particularly to the east and north-east. At first glance, this appears to support the narrative of outward family displacement.

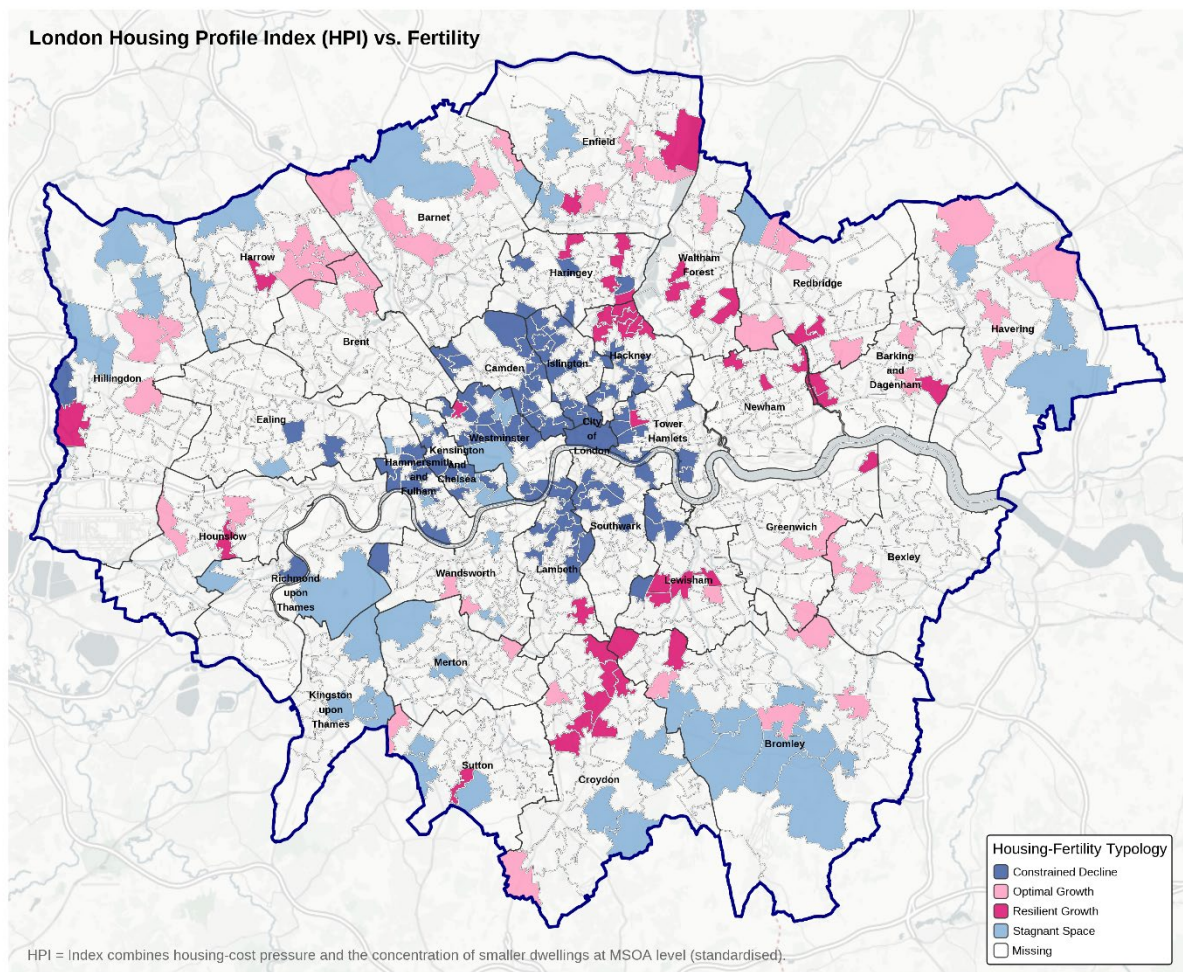


But if we look at city-wide averages, they mask important variation. The relationship between fertility and housing pressure is weak at a metropolitan scale. Some

high-pressure areas, i.e. high HPI, have low fertility; others do not. This suggests that housing pressure does not operate uniformly across London.

Within boroughs, meaning if we look at the context, the variation is different. In places such as Hackney and Newham, housing pressure explains neighborhood-level fertility differences more effectively than a wide overview, underscoring the importance of local context.

Beyond Averages: Four Neighborhood Types



To understand the uneven relationship between housing pressure and fertility, London's neighborhoods can be grouped into four broad types:

Constraint Decline: High housing pressure and low fertility.

Concentrated in the inner core, these neighborhoods align most closely with the conventional urban form of mixed-use areas with high concentrations of commercial and office space.

Optimal Growth: Low housing pressure and high fertility.

Predominantly, outer-London areas where larger dwellings and lower costs coincide with higher birth rates and align with the narrative of inner migration.

Stagnant Space: Low housing pressure but persistently low fertility.

Often, transitional or ageing neighborhoods are where fertility is shaped by factors beyond housing.

Resilient Growth: High housing pressure and high fertility.

The most revealing category. These neighborhoods demonstrate that family formation can persist even under severe structural constraints.

Where Fertility Persists

Neighborhoods where families persist despite constraints cluster in East and North-East London, particularly Hackney, Haringey, Lewisham, Waltham Forest, and Newham. Here, high housing costs and many small dwellings coexist with fertility rates above the city average.

Hackney's example shows that this persistence is driven not only by housing, but also by demographics and culture: a large Charedi Jewish population with high fertility, diverse ethnic groups with above-average birth rates, a young population, and ongoing inward migration all boost fertility. These factors sustain high birth rates despite structural constraints. [@Hackney_Demo]

Importantly, this is not a city-wide pattern: even within boroughs, some neighborhoods show resilience while others do not. Fertility resilience is a local, not an administrative, phenomenon. Families are not universally displaced from high-cost, high-constraint areas; in some localities, they continue to form and maintain households despite these challenges, though for others, constraints may eventually prevent family formation. [@No_babies]

Conclusion

Housing pressure does not consistently hinder family formation in London. While high costs and limited dwelling sizes are often assumed to push families outward at some point, their influence is highly context-dependent. Citywide, the relationship is weak; within boroughs, it becomes clearer.

Most importantly, the analysis identifies neighborhoods where family formation persists despite severe housing constraints and overlooked non-structural factors. These clusters challenge the assumptions of displacement. Rather than simply losing families to peripheries, London is reorganizing where family life remains possible or rather, which areas are the last strongholds.

Recognizing that borough context might shape neighborhood fertility patterns has clear implications for future analysis and policy. Any effort by the administration to support early family formation must look beyond housing metrics and consider the broader characteristics that make neighborhoods viable for young families. These include demographic composition, cultural networks, access to schools, open space, childcare, and local amenities — factors not fully captured in this report.

Equally important are the dynamics of internal and international migration, which influence who enters, leaves, and remains in those neighborhoods. Any future in-depth research should incorporate these flows to better understand the conditions under which families choose to stay, settle, or move. Only by recognizing the full ecology of family functionality, spatial, social, and economic. Afterwards, we can develop a more nuanced understanding of urban life.

Technical Appendix: Methodological Framework

Data, Methods, and Modelling Strategy

1. Data Sources and Preparation

The following section details the data, methods, and modelling strategy used to explain the preceding analysis. This study integrates multiple open datasets at the Middle Super Output Area (MSOA) [MSOA_LINKS] level to examine the relationship between housing conditions and fertility patterns across London.

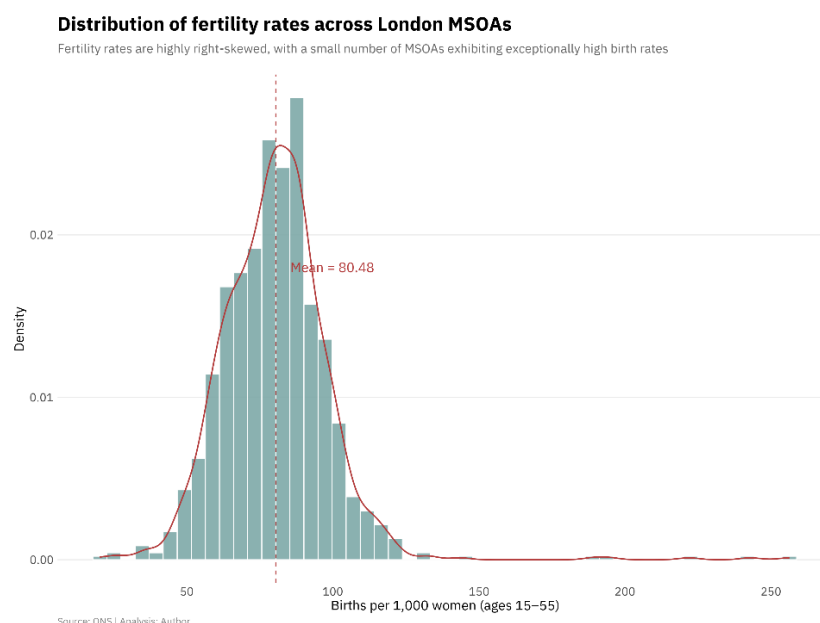
1.1 Births, Female population and Fertility

Birth data were obtained from the Greater London Authority's Births and Fertility Rates dataset. Birth records were aggregated over 2019–2021 to reduce annual inconsistency and smooth short-term fluctuations. Births were grouped by maternal age bands and summed for women aged 15–55, yielding cumulative births per MSOA over the three-year period. [Birth_rate]

Population denominators were derived from a female population dataset [Data_pop], filtered to the same age range (15–55) and aggregated to the MSOA level. Fertility rates were then calculated as:

$$\text{Fertility Rate} = \frac{\sum \text{Births}_{2019-2021}}{\sum \text{Female Population}_{15-55}} \times 1000$$

This produces a standardized measure of births per 1,000 women of childbearing age.



1.2 Housing Form: Dwelling Size Structure

Dwelling data were sourced from the London Dwellings by Property Type dataset (2016), reported at the MSOA level. Dwellings were grouped by number of bedrooms, and records of missing values were excluded. To approximate constraints on family-sized housing, the analysis focused on the proportion of small dwellings (one-bedroom units) relative to larger ones (two-bedroom or more). Counts were converted into MSOA-level percentages. **[@Data_Housing]**

1.3 Housing Costs

Income data were drawn from the Office for National Statistics Small Area Model-Based Income Estimates (Financial Year Ending 2023). Housing costs were operationalized as the difference between disposable (net) annual income before housing costs and disposable (net) annual income after housing costs. This measure captures financial pressure from housing costs per household but does not distinguish tenure type or household size. **[@Data_H_Price]**

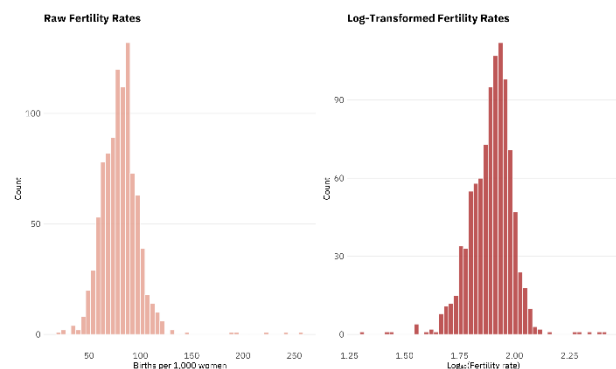
2. Variable Transformation and Correlation Structure

All continuous variables were transformed to improve normality and comparability:

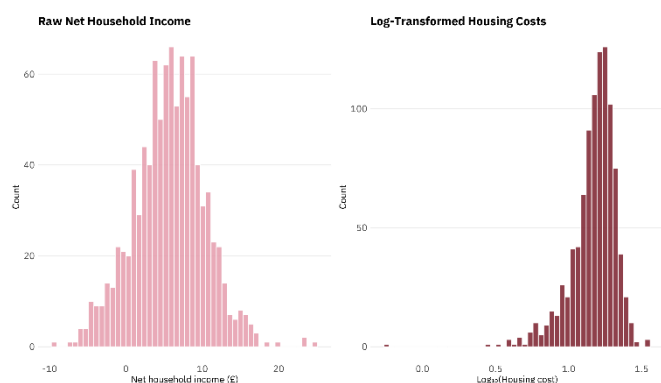
Fertility Rate ($\log_{10}$ transformed):

To address the extreme positive skew and heavy right tail in the raw MSOA data.

Effect of log transformation on fertility rates



Distributional effects of log transformation on income and housing costs



Small Dwellings (Logit-

transformed): Applied to the proportion of small dwellings to map bounded percentage data (0–1) onto an infinite scale.

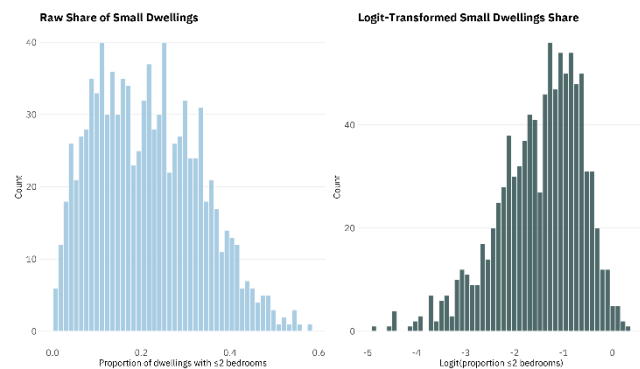
Housing

Costs($\log_{10}$ transformed):

Applied to stabilize the variance

across the range of predictors, even where the raw income distribution appeared relatively symmetrical.

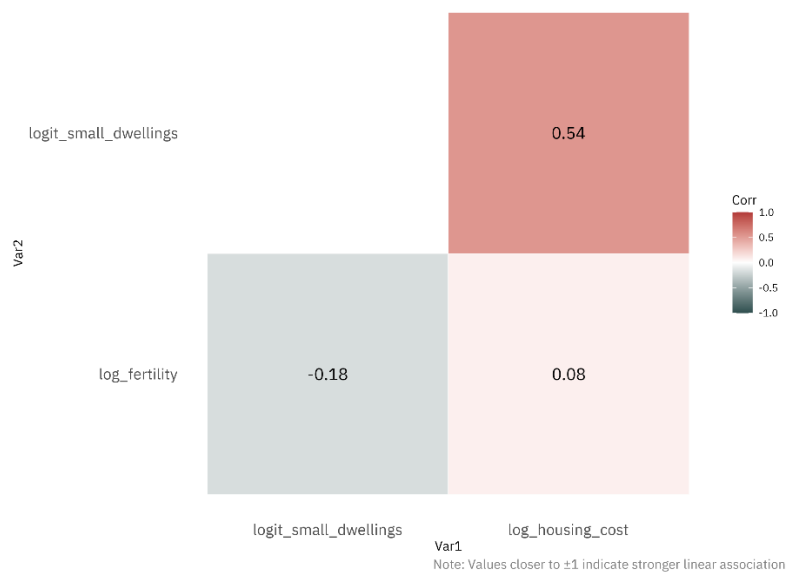
Distribution of small-dwelling prevalence before and after logit transformation



Pairwise Pearson correlations indicated a moderate positive correlation between housing costs and small-dwelling prevalence ($r = 0.54$), but only weak associations between individual housing variables and fertility. This motivated me to approach it through PCA.

Relationships between fertility, housing costs and dwelling size

Pairwise Pearson correlations after log and logit transformations



3. Construction of the Housing Portfolio Index (HPI)

To capture the multidimensional nature of housing pressure, Principal Component Analysis (PCA) was applied to the logit-transformed proportion of small dwellings and log-transformed housing costs. The first principal component (PC1) explained most of the shared variance, with both variables loading equally at 0.707.

$$\text{PC1 (HPI)} = 0.707 \cdot \text{logit}(\text{small dwellings}) + 0.707 \cdot \log(\text{housing cost})$$

This component was retained as the **Housing Portfolio Index (HPI)**. High HPI values characterize neighborhoods facing the dual constraint of high housing financial costs and restricted physical space (less than 2 bedrooms). Initial validation via a city-wide Pearson correlation revealed a weak negative association between the HPI and log-fertility ($r \approx -0.06$), suggesting that housing constraints do not exert a uniform linear influence across the capital, prompting me to conduct an ANOVA test.

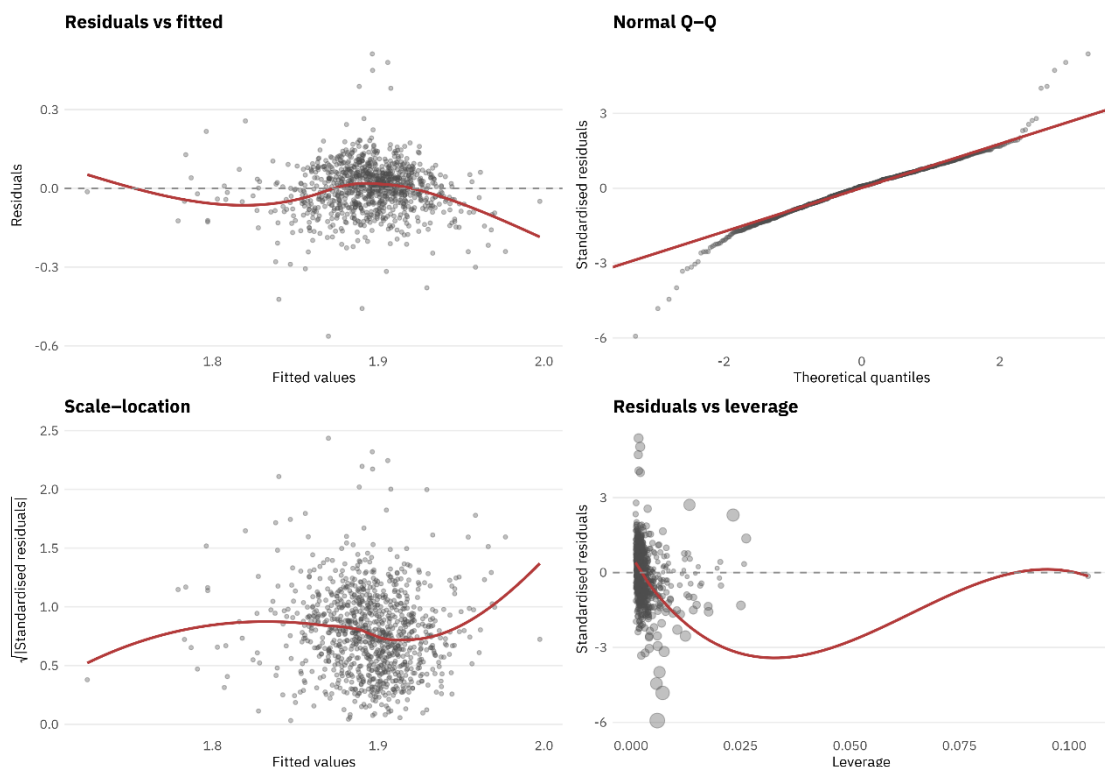
4. Inferential Modelling

4.1 Linear Regression

Baseline linear models showed that the negative association of small-dwelling prevalence (-0.031) was stronger than the positive association of housing cost (0.024). Despite statistical significance, explanatory power remained low ($R^2 \approx 0.07$). This suggests that the real drivers of fertility lie in unmeasured local factors, such as community support and cultural networks within the borough structure.

Linear model diagnostics

`log_fertility ~ logit_small_dwellings + log_housing_cost`



4.2 Borough-Level Effects (ANOVA)

A one-way ANOVA proved that birth rates vary drastically depending on which neighborhood is in ($F = 17.02, p < 2 \times 10^{-16}$). This high level of variation confirmed

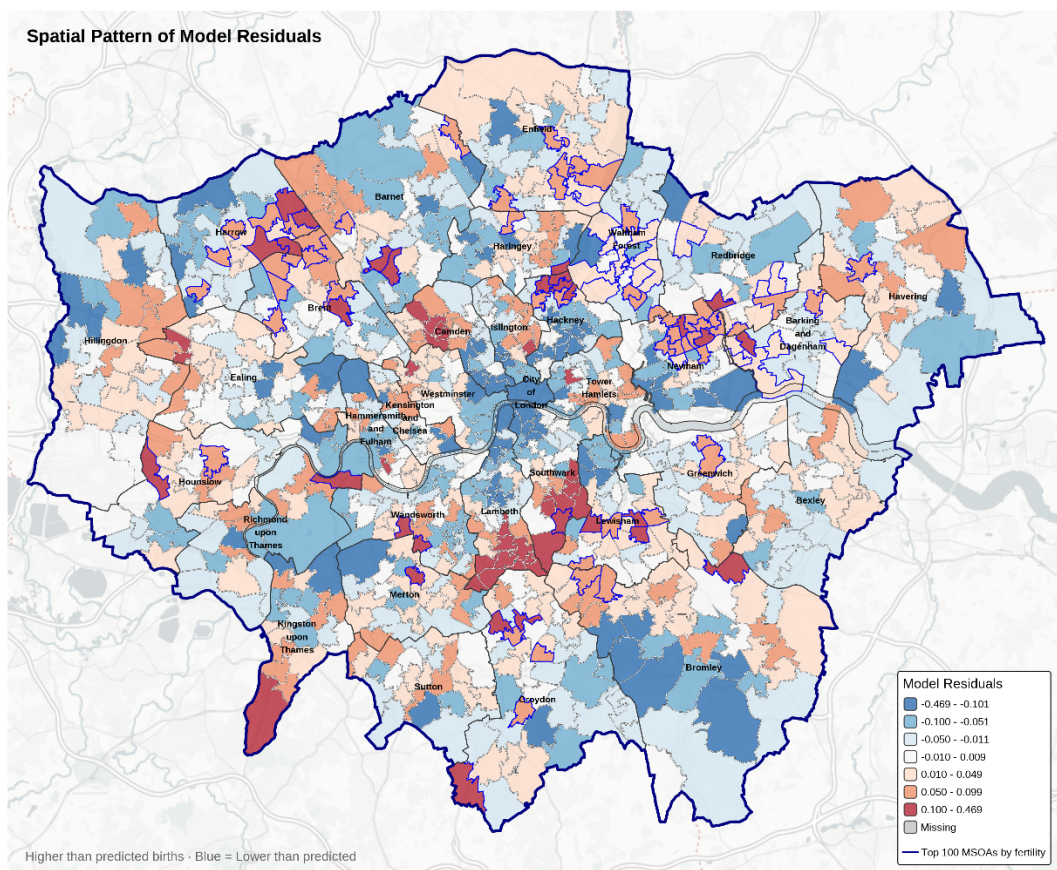
that a simple city-wide model was insufficient and required a more sophisticated mixed-effects approach that accounts for borough-level structures.

4.3 Linear Mixed-Effects (LME)

MSOAs were nested within boroughs to account for the hierarchical structure of London's administrative and social geography. The final Analysis was:

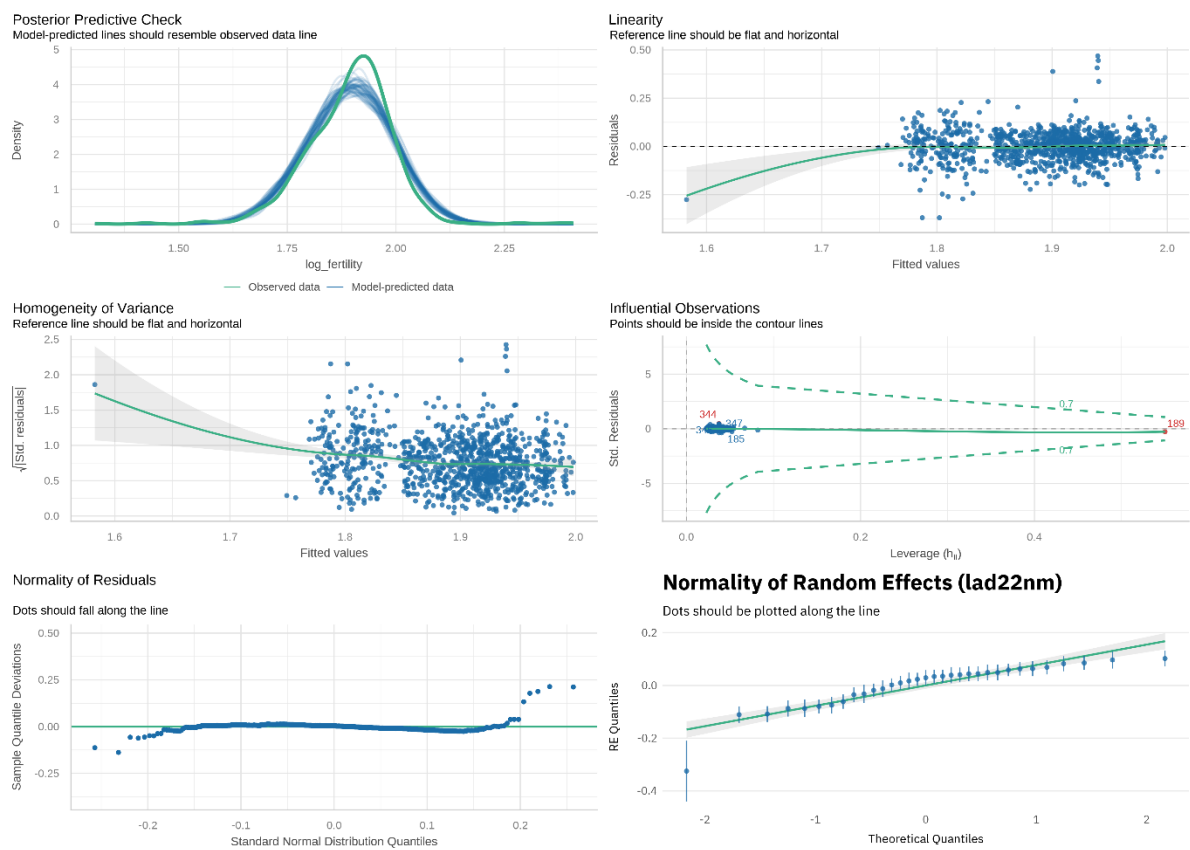
$$\log(\text{Fertility}) \sim \text{HPI} + (1 \mid \text{Borough})$$

The model confirms that the **Housing Portfolio Index** has a clear, statistically significant impact on birth rates ($p < 0.001$). However, the variation caused by **borough** structure (0.0075) is greater than that caused by individual neighborhood MSOA(0.0064). This means that the borough a family lives in and the social support it provides are just as critical to fertility as the physical size or cost of their home.



Diagnostics identified four high-zone MSOAs in Hackney, Camden, and the City of London. The Hackney outliers recorded the dataset's highest fertility rates ($\log_{10}$ > 2.38), disproportionately affecting the borough's random intercept. These were

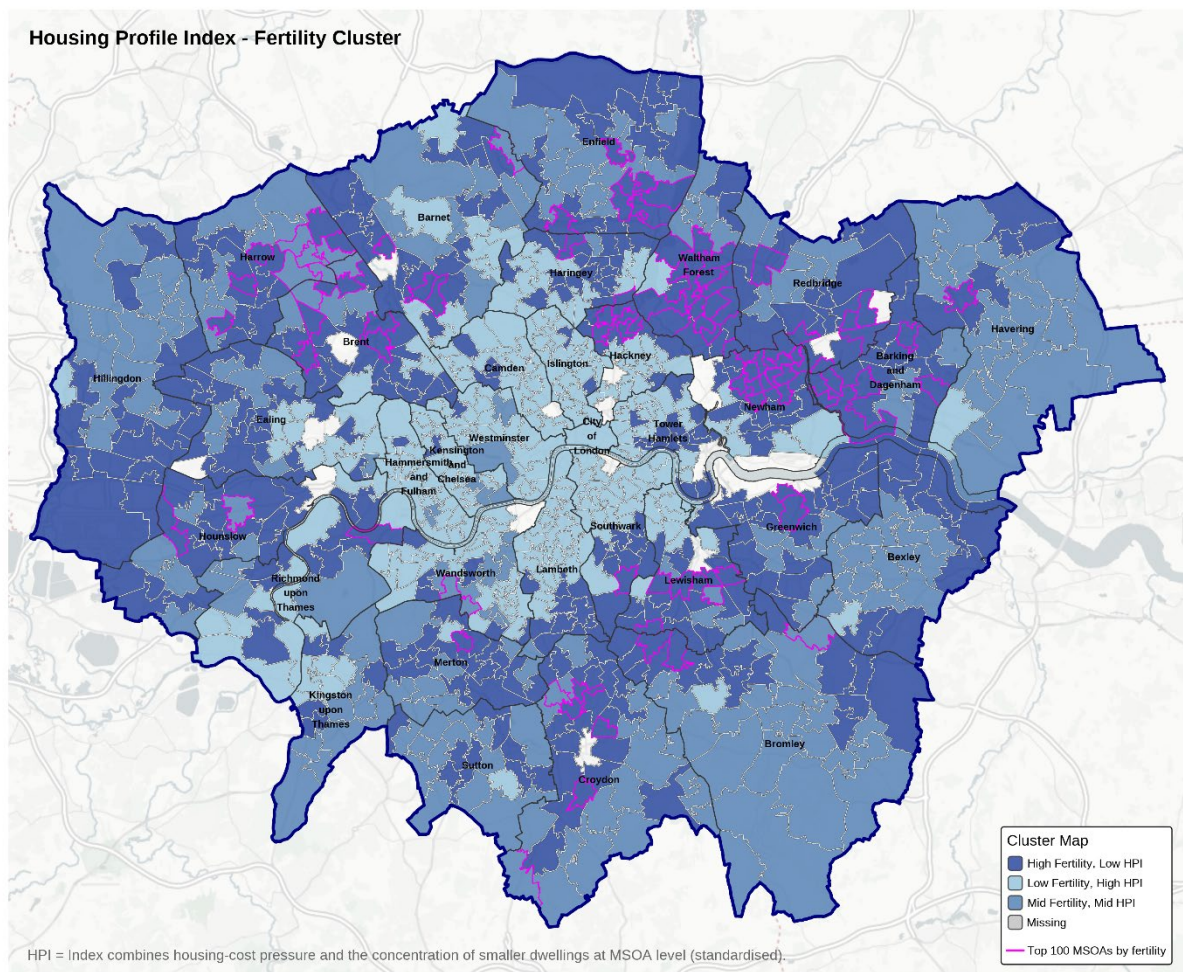
retained to preserve the model's spatial integrity and represent genuine demographic "extreme zones". Robustness is confirmed by the **Posterior Predictive Check**, where the predicted data closely tracks the observed distributions, and by the **Normality of Random Effects** plot, which verifies hierarchical stability.



5. Cluster Analysis and Typology

K-means clustering was applied to standardized fertility, dwelling size, and dwelling cost data. A 3-cluster solution was selected based on the elbow method:

- **Cluster 1 (n=454):** Balanced baseline.
- **Cluster 2 (n=231):** High rooms, lower cost.
- **Cluster 3 (n=278):** Minimum rooms, highest cost areas.



Beyond broad clustering, a more selective typology map was needed to better understand London's existing structure.

Categories are defined by quantile thresholds of fertility and the Housing Portfolio Index (HPI):

- **Resilient Growth (n=47):** High fertility (top 25%) despite high HPI constraints (top 25%); concentrated in **Hackney** (7), **Haringey** (5), and **Newham** (3).

REFFFF